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Short Contribution

Step-Like Structure in Temperature and Salinity Profiles, Observed near Icebergs Trapped by Fast Ice, Antarctica

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Step-like thermohaline structure was observed near icebergs trapped by fast ice in Lützow-Holm Bay, Antarctica. The structure was developed within the draft depth of the icebergs. The typical vertical distance between the succeeding step surfaces is 20–30 m, and the temperature and salinity differences across the step surface are about 0.05–0.06 deg and 0.06–0.10 psu, respectively. The structure appears to be generated by a sea-water/iceberg interaction, and is explainable by a simple model in which an ice wall is placed in the ocean linearly stratified by vertical salinity gradient.

1. Introduction

Icebergs are formed when glaciers flow out over land into the adjoining sea and break off or calve. Melting of icebergs would influence the oceanic structure and current system nearby, by extracting latent heat and by supplying fresh water. Melting of icebergs takes place preferentially along the sides rather than the top or bottom (Huppert and Turner, 1980).

There had been a series of laboratory investigations on ice walls melting into surrounding water. In the uniform salinity case, along the vertical ice walls, a turbulent boundary layer with solely upward motion forms (Josberger and Martin, 1981). In the case of the salinity gradient, however, flow in the vicinity of the ice is very different (Huppert and Turner, 1980; Huppert and Josberger, 1980). Next to the ice there is a thin upward-flowing layer of meltwater, beyond which there is a thicker, turbulent, downward-flowing layer. This outer turbulent layer entrains meltwater from the inner boundary layer. Because of the increase in salinity, and thereby density with depth, the water in this cooled outer layer reaches a level of neutral buoyancy and flows away from the ice in a series of nearly horizontal layers.

On the other hand, there have been few field observations carried out near icebergs or at the terminus of a tidewater glacier. Jacobs *et al.* (1981) observed a step-like thermohaline structure with a conductivity-temperature-depth (CTD) cast taken near the Erebus Glacier Tongue, McMurdo Sound, Antarctica. They inferred that the structure is caused by the ice walls melting into vertically stratified salt water, analogous to the laboratory experiments of Huppert and Turner (1980). Josberger and Neshyba (1980) observed step-like temperature structure around an iceberg off Newfoundland with an expendable bathythermograph (XBT) survey. Allison *et al.* (1986) made CTD observations at various close distances around two icebergs during Australian Antarctic Division cruises. Although they found evidence of modification of the near surface water in the vicinity of the iceberg, no strong evidence was found of step-like structure

derived from ice walls melting.

Under the Japanese Antarctic Climate Research (ACR) program, extensive observations of ocean and sea ice have been carried out in and off Lützow-Holm Bay during 1990–1992, by the 31st and 32nd Japanese Antarctic Research Expedition (Takizawa *et al.*, 1992; Ohshima *et al.*, 1993). As a part of the program, we carried out several oceanographic observations as close as possible to an iceberg trapped by the sea ice in Ongul Strait, which is located in the eastern part of Lützow-Holm Bay, to examine the influence of the iceberg melting in the ocean.

Lützow-Holm Bay is usually covered with fast ice, thus the icebergs are trapped by the sea ice as long as the fast ice does not break off, which allowed us to work close to the iceberg safely. Furthermore, because the fast ice cover prevents the atmosphere from disturbing the ocean, the effect of the iceberg melting can be extracted from the observations.

2. Observational Sites and Method

An isolated iceberg, which is located in the center of Ongul Strait and is trapped by fast ice, is selected for our investigation. Figure 1 is an air photograph of the selected iceberg, indicated by the arrow, and the surrounding geography. The iceberg is located in the center of Ongul Strait, trapped by fast ice. Figure 2 shows a photograph of the iceberg. The length of the iceberg is approximately 100 m on the water line and its freeboard height is about 10–15 m. The water depth at the site is about 440 m.

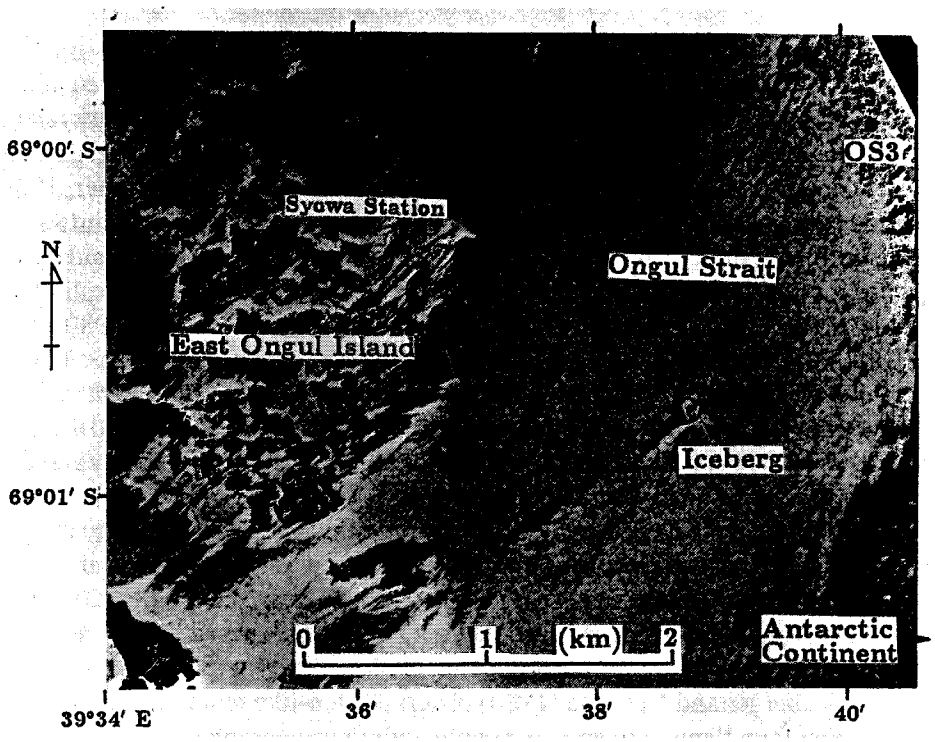

Fig. 1. An air photograph showing the location of the observed iceberg, denoted by the arrow, on December 6, 1991. At the station OS3, long-term moored current measurements were made from April to December, 1991 (photograph courtesy of S. Nakajima, Geographical Survey Institute).

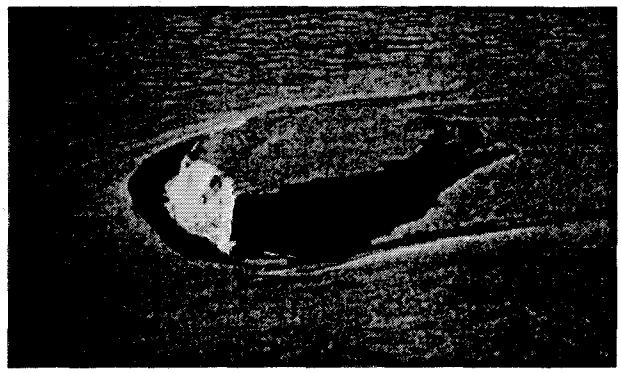

Fig. 2. An air photograph of the observed iceberg, taken from the north of the iceberg, on January 24, 1992.

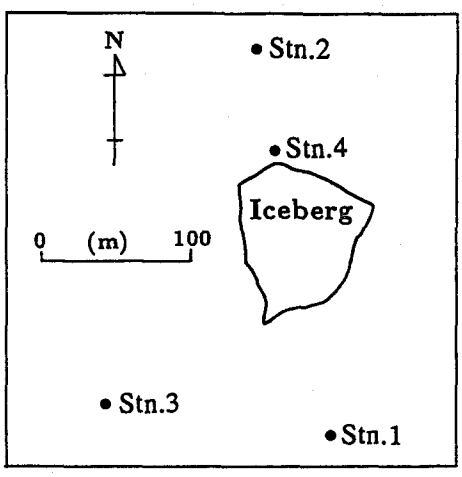

Fig. 3. Locations of the observational stations around the iceberg.

The observations with CTD (Seabird: SBE-19) were made in two seasons: fall (May) and early summer (November and December) in 1991. The fast ice in Ongul Strait has not broken off since 1989. Thus, during the observations, the iceberg was trapped by the fast ice. Figure 3 shows the location of the oceanographic stations.

At the station OS3, located about 2 km north-eastward from the iceberg (see Fig. 1), a currentmeter, which was moored to the fast ice, made long-term measurements at depth of 70 m from April to December 1991. According to the current measurements, the direction of the mean current changed from southward to northward in May (Ohshima *et al.*, 1993). Since the Ongul Strait is narrow and its bottom contours run nearly straight along the north-south direction, the current at Stn. OS3 is expected to be similar to that at the iceberg and used as the reference.

3. Results

Figures 4(a) and (b) show the CTD profiles taken at Stn. 2 and Stn. 3 on December 2, respectively. As shown in Fig. 4(c), the current direction is northward at the time of CTD

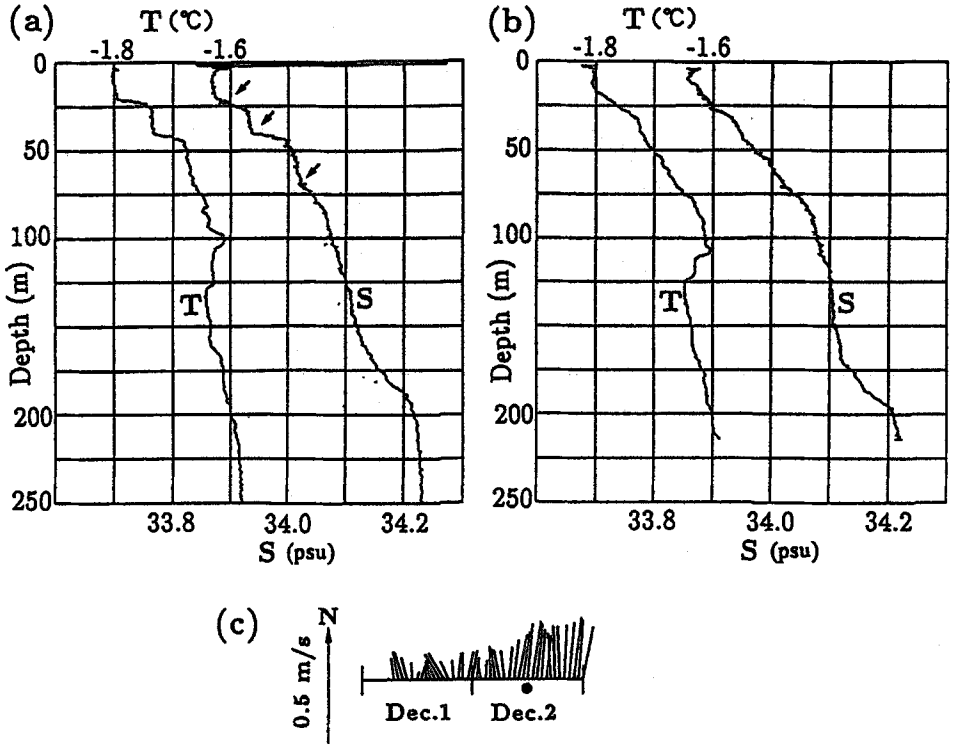

Fig. 4. (a) Temperature and salinity profiles of CTD cast at Stn. 2 on December 2, 1991. (b) As in (a) but at Stn. 3. (c) Stick diagram of the current at Stn. OS3 during December 1–2. The black circle shows the time of the CTD observation.

observation. Stn. 2 is located downstream of the iceberg while Stn. 3 upstream. Both of the temperature and salinity profiles at Stn. 2 show clear step-like structure, as indicated by the arrows in Fig. 4(a). The density shows similar profile to the salinity, since density depends strongly on salinity at low temperatures. The stepped structure occurs to depths of 75 m from the sea surface. The upper two steps are distinct while the lowest one is not clear. The vertical distance between the succeeding step surfaces is about 20 m, and the temperature and salinity differences across the step surface are about 0.06 deg and 0.06 psu, respectively. On the other hand, at Stn. 3, which is located upstream of the iceberg, the profiles do not show such features (Fig. 4(b)).

At Stn. 2, CTD observations were carried out also on November 18 and December 11 (Fig. 5). In the case of November 18 (Fig. 5(a)), several steps are found to depths of 80 m from the sea surface. The scale of the steps appears to be relatively small at the depths from 25 to 50 m. In the case of December 11 (Fig. 5(c)), however, no distinct steps are seen. Therefore, out of three observations, two observations show clear step-like feature at Stn. 2. On December 11, the observation was made also at Stn. 4, located at the distance of only 7 m from the iceberg (see Fig. 3), and the results are shown in Fig. 5(d). There is slight suggestion of step-like features of smaller scale (at the depths from 30 to 60 m in the salinity profile), though they are not so significant. Note that Stn. 2 and Stn. 4 are considered to be located downstream of the iceberg during the observations, since the current direction is northward in this season (Ohshima *et al.*, 1993; see

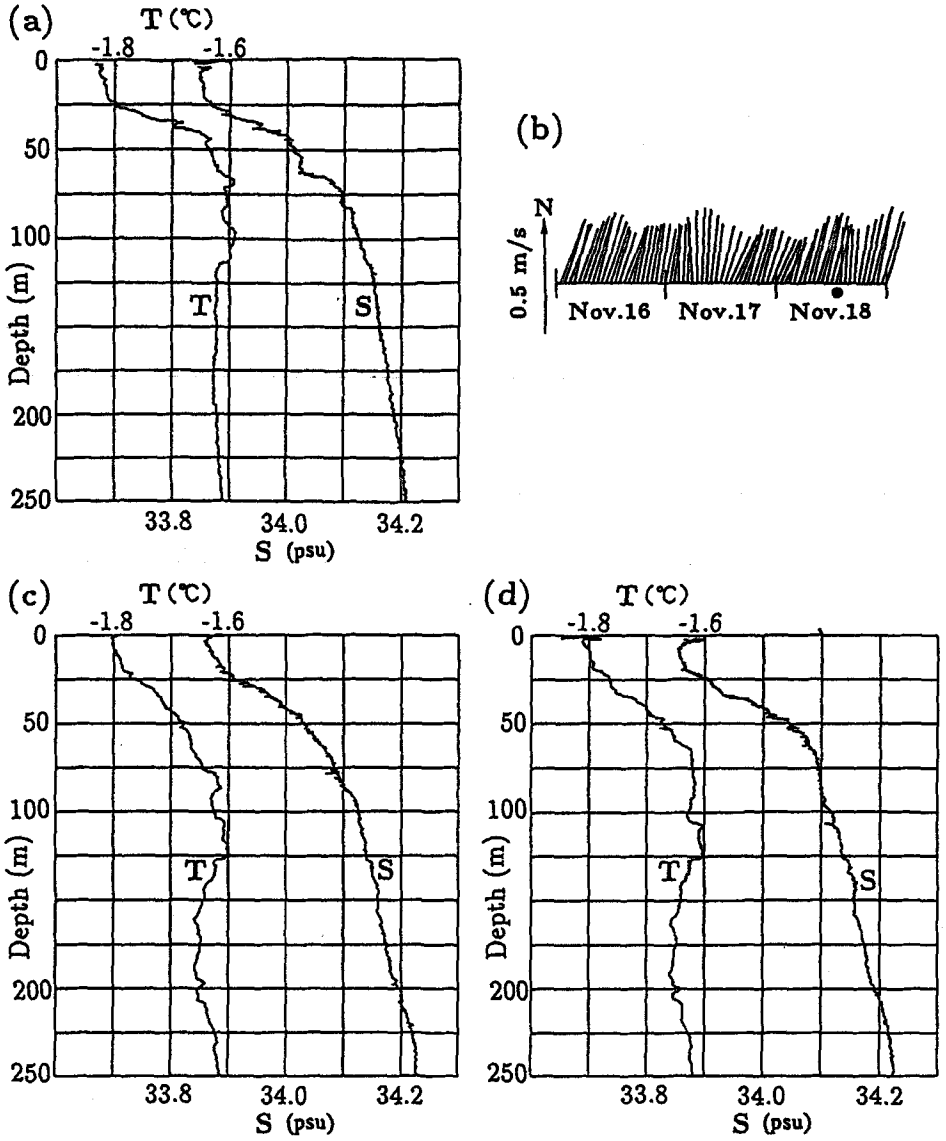

Fig. 5. (a) Temperature and salinity profiles at Stn. 2 on November 18, 1991. (b) Stick diagram of the current at Stn. OS3 during November 16–18. The black circle shows the observation time. (c) As in (a) but on December 11, 1991. (d) As in (c) but at Stn. 4.

also Fig. 5(b)).

Judging from the freeboard height of the iceberg, 10–15 m, the draft depth of the iceberg is estimated to be about 70–100 m. This seems to correspond to the observed depth range of the stepped features at Stn. 2. In this season, water temperature in the upper layer is above the in-situ freezing point by about 0.1–0.3 deg, as shown in Figs. 4 and 5.

In May, we made CTD observation at Stn. 1. The site is located downstream of the iceberg, since the current direction is southward at this time as shown in Fig. 6(c). The results of Stn. 1 are shown in Fig. 6(a). The profiles of temperature and salinity do not show step-like features.

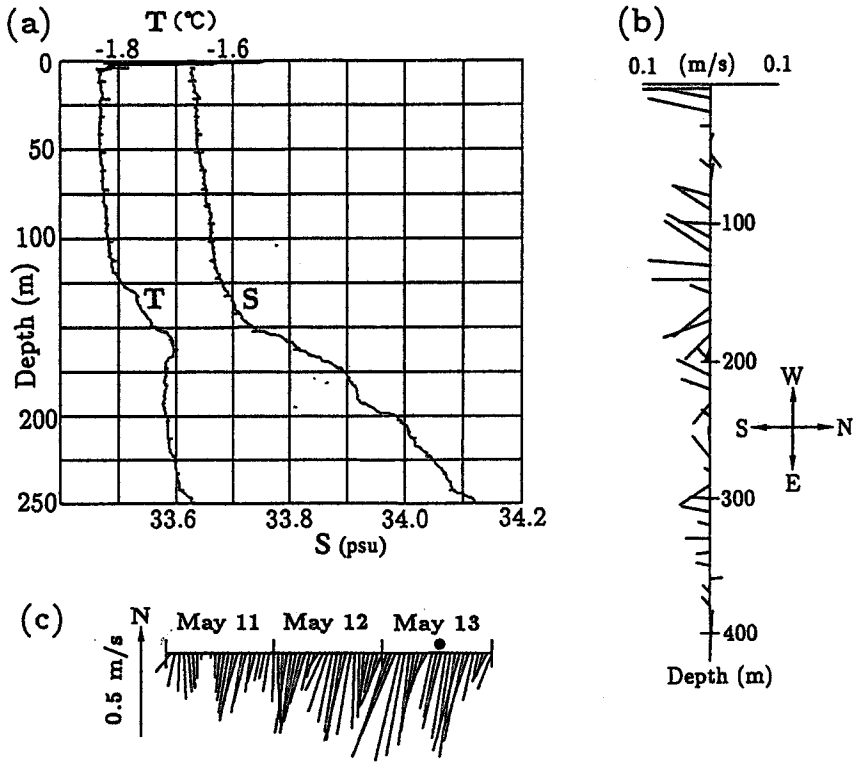

Fig. 6. (a) Temperature and salinity profiles at Stn. 1 on May 13, 1991. (b) Vertical profile of horizontal velocity measured with the electro-magnetic currentmeter at Stn. 1. (c) Stick diagram of the current at Stn. OS3 during May 11–13. The black circle shows the observation time.

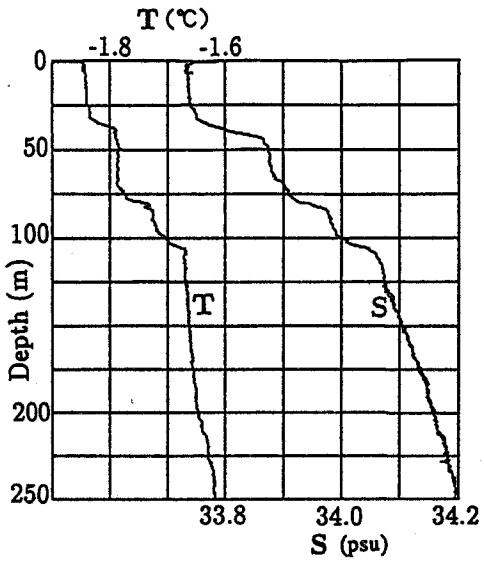

Fig. 7. Temperature and salinity profiles near an iceberg located on 69°15.8′ S, 39°29.8′ E. The observation was made on August 25, 1991.

Note that in this season water temperature in the upper layer is very close to the freezing point, within 0.05 deg. Figure 6(b) shows the vertical current profile observed with an electro-magnetic current meter. The current is quite small at the depths from 30 to 70 m, which may be due to the shelter effects of the iceberg.

Clear step-like structure was also observed near a different larger iceberg in August (Fig. 7). This iceberg, trapped by fast ice, is located on 69°15.8′ S, 39°29.8′ E, about 30 km south from the Syowa Station. The vertical distance between the succeeding step surfaces is about 30 m, and the temperature and salinity differences across the step surface are about 0.05 deg and 0.10 psu, respectively. The features extend to depths of 100 m from the surface. This depth seems to correspond to the draft depth of the iceberg, judging from the freeboard height of the iceberg.

During the ACR program, we made a total of 115 CTD casts, at 30 stations under fast ice in Lützow-Holm Bay. Almost all the stations except the ones presented above are located far, more than several kilometers, from icebergs, and clear step-like features can not be seen in the upper layer.

4. Discussion

The stepped structure has been observed only on stations which are close to an iceberg. Further, the steps are developed within the draft depth of the iceberg. These suggest that this structure is generated by a sea water/iceberg interaction. More specifically, the step structure or the layering is explainable by a simple model in which an ice wall is placed in the ocean linearly stratified by vertical salinity gradient, analogous to the laboratory experiments of Huppert and Turner (1980).

In the real ocean, however, ideal conditions produced in the laboratory experiments are hardly achieved. Even if the layering is generated by the ice walls melting in the ocean, it can be soon destroyed by the atmospheric and/or oceanic disturbances. While in the presence of fast ice, the atmospheric disturbances are prevented. In the Ongul Strait, the mean current exists with a typical velocity being 0.3 m/sec. The region upstream of the iceberg would be hardly affected by the iceberg. On the other hand, the downstream region just near the iceberg is probably stagnant owing to the shelter effects of the iceberg, as suggested in Fig. 6(b). Such conditions allow the layering to be maintained. These are consistent with the fact that the steps were observed only downstream of the iceberg in the case of Dec. 2 (Fig. 4). With increase of the upper layer temperature, the ice wall melting, accordingly the layering would be more active. This is consistent with the fact that the steps were found only in early summer (Figs. 4 and 5), but not in fall when the upper layer temperature is very close to the in-situ freezing point (Fig. 6).

The laboratory experiments of Huppert and Turner (1980) and Huppert and Josberger (1980) and their related analyses suggest that the layer scale h is given by

$$h = 0.65\left[\rho\left(T_f, \overline{S}_\infty\right) - \rho\left(\overline{T}_\infty, \overline{S}_\infty\right)\right] \Big/ \frac{d\rho_s}{dz}$$

where $\rho(T, S)$ is the density as a function of temperature and salinity, T_f the freezing point of water at the mean far-field salinity $\overline{S}_\infty$, $\overline{T}_\infty$ the mean far-field temperature and $d\rho_s/dz$ the vertical density gradient due to salt. $d\rho_s/dz$ is evaluated by fitting a line of constant slope through the salinity profile. The layer thickness predicted from the equation is 1.6 m and 1.2 m for Fig. 4(a) and Fig. 7, respectively. The observed major thickness is one order larger than the predicted one. Also in the case of the Erebus Glacier, Jacobs *et al.* (1981) showed that the observed layer thickness is

larger than the predicted one by a factor of 2 or more. One possible mechanism which explains the discrepancy between the observations and prediction is merging of the layers: the temperature and salinity differences across the layer interface decrease to zero, or the interface migrates vertically to the adjacent interface. Even if the vertical scale of the steps is small initially, it may become larger gradually with merging. Ambiguous smaller scale steps seen in Figs. 5(a) and (d) might be the remnant of the initial steps.

At the present stage, however, the observational data is too limited to conclude that the layering mechanism is similar to that of the laboratory experiment and that the merging actually occur. Precise understandings of the layering mechanism must await more-detailed field observations in the future.

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